Note: Need to be aware of basic procedures for determination of latent heat.

2 (a) (i) The tension in the rod alone is not sufficient to provide the centripetal force required to keep the ball in circular motion. The balance has to be provided by the gravitational force acting on the ball (mg).

Note: Candidates must realise that this is not an equilibrium situation.

(ii) The centripetal force required is the resultant force,

$T + mg = 2mg + mg = 3mg$

(iii) When ball is below C, the tension T and mg are in opposite directions.

$T - mg = 3mg$

$T = 4mg$

(b) (i) Centripetal force, $mr\omega^2 = 3mg$.

$\omega = 6.4\,\text{rad}\,\text{s}^{-1}$

(ii) Linear speed, $v = r\omega = (0.72)(6.393) = 4.6\,\text{m}\,\text{s}^{-1}$

(c) (i) The rigid rod has to exert an upward force on the ball against gravity, to prevent this loss in gravitational potential energy from transforming into gain in kinetic energy. Work has to be done. Since the direction of the exerted force is opposite to displacement, work done is negative.

Note: Common misconception was to state work is done because centripetal force has changed direction.

(ii) Taking downwards as positive

$$\text{Work done}, W = Fd = (-mg)(h)$$
$$= (-0.240 \times 9.81)(2 \times 0.72) = -3.4\ \text{J}$$

3 (a) (i) A field of force is a region of space where a body experiences a force due to the